Predicting Patient Diagnosis of Osteoporosis using CT Scan Images with Machine Learning: A Study using Pixel Density Counts to Determine Bone Mineral Densities and T-scores

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Abstract

Purpose
Osteoporosis develops when there is a decrease in bone mineral density or bone mass. Therefore, it is important to accurately predict the bone mineral density of patients to understand the health of their bones. The primary purpose of this research was to help determine osteoporosis by automating bone mineral density assessments, using computer imaging and machine learning techniques.

Common methods used to measure bone density such as DEXA scans, are limited and often involve labor-intensive procedures that are susceptible to human error. This new strategy aims to improve osteoporosis treatment and treatment outcomes by using CT scans, which are three-dimensional, to increase information and diagnostic efficiency and accuracy.

Methods
CT scan images from 30 different patients were analyzed. Patient lower lumbar spines were intentionally analyzed due to its importance in current osteoporosis screenings. Using a dataset of segmented and labeled CT scans, the algorithm processed the scans to calculate each patient's bone mineral density and T-score. Bone mineral density values obtained were compared with figures in existing journals, since features of the dataset and patient information were unknown.

Results
The bone densities and T-scores of each patient were calculated. Using T-scores, patients were diagnosed with the following bone conditions: normal, osteopenia, or osteoporosis. Two out of the thirty patients (0.067%) were diagnosed with osteopenia, while the remaining were determined to have normal bones. The average bone density among segments L1, L2, L3, and L4 were 1.8397, 1.9017, 1.8671, and 1.8715 g/ cm^2, respectively, and the averaged L1-L4 bone density was 1.8715 g/ cm^2. Analyzing the comparison of segments L1-L4, in most cases, L2 and L4 have comparatively higher BMD values than L3 and L1.

Conclusion
The study demonstrated that pixel density counts could be used in machine learning to determine bone densities. Bone density measurements can assist with the determination of T-scores and osteoporosis diagnoses. With further research and more accessibility to patient information, assumptions can be fine-tuned, and model predictions can be compared with DEXA scan results using the same dataset.

1. Introduction

1.1. What are you trying to do? Articulate your objectives using absolutely no jargon.

Osteoporosis is a bone disease that develops when a person's bone mineral density decreases beyond a globally established threshold. It is also known as a "silent" killer because there are typically no associate symptoms. A decrease in bone mineral density can increase the risk of fractures (PLAN, L. R., & Areas, N. S. S. (2010)). The most common way to diagnose a person with osteoporosis is by determining their T-score, the classification system for bone density established by the World Health Organization (WHO) (Kanis, 1994). Determining a patient's T-score also aids in determining the likelihood of fracture.

To aid in this silent epidemic, our goal is to develop a tool that uses advanced technology to analyze CT scans of the spine, which include vertebral labeling and segmentation, to determine the bone mineral density (BMD) and the likelihood of fractures of each segment without the need for manual measurements. Our research question is, using image pixel density analysis, can a machine-learning algorithm predict the bone mineral density of a patient's spine segments to determine the health of their bones? This tool will aim to improve patient care by enabling personalized treatment plans based on precise assessments of bone health.

1.2. How is it done today, and what are the limits of current practice?

Currently, BMD assessments are predominantly conducted through dual-energy X-ray absorptiometry (DEXA or DXA) scans (National Institute of Arthritis and Musculoskeletal and Skin Diseases, 2023), which have limitations due to their two-dimensional nature. This can lead to inaccuracies in readings, potentially affecting diagnosis and treatment plans. Medical experts also manually analyze these scans, a process that can be time-consuming and subject to human error.

1.3. What is new in your approach and why do you think it will be successful?

Our approach introduces a novel method for assessing bone density using segmented CT scans and machine learning (ML). Unlike DEXA scans, CT scans are three-dimensional which allows for additional information during readings. This tool will be able to read the segmented CT scans of the spine then use machine learning to measure bone density. By automating the process of identifying bone density and the likelihood of fractures in spine segments, we aim to significantly reduce the time, effort, and potential inaccuracies associated with manual measurements. This innovation is crucial, especially for patients undergoing radiation therapy for various cancers, where bone health can be compromised. Because of radiation, various bones are innocent bystanders of the treatments, resulting in some becoming fractured. We believe our approach will succeed because it combines advanced technology with clinical needs, offering a more accurate and efficient solution for diagnosing osteoporosis and other spine-related conditions.

1.4. Who cares? If you are successful, what difference will it make?

If successful, our research could help transform the medical community and how bone density is measured, by offering a more accurate, efficient alternative to current methods. It could particularly impact the field of osteoporosis management by allowing for earlier and more precise detection, which could lead to better patient outcomes.

1.5. How will you measure success?

Success will be measured by the determination of bone mineral density and the machine-learning model's ability to detect if a patient has normal/healthy bones, osteopenia, or osteoporosis based on the labeled and segmented CT scan data. For this assignment, we will assess the model's performance through a rigorous validation process by comparing its predictions with manual measurements and expert diagnoses of patients who have osteoporosis and patients who do not through literature and research to determine if the results are adequate. Researchers have used Artificial Neural Networks and image analyses using deep learning in this field as another way to determine bone density. In the future, we would like to compare our model's results with results from other ML models using the same dataset.

2. Literature Review

Identifying individuals with low bone mineral density (BMD) is crucial for preventing and treating osteoporosis, a condition characterized by reduced bone strength and increased fracture risk (Raisz L. G. 2005). Bone strength depends on both BMD, which reflects bone mass, and bone quality, including structure and microfractures. Given the asymptomatic nature of bone loss and the challenges in treating osteoporosis pharmacologically without early intervention, early detection of low BMD individuals is critical for effective osteoporosis management and intervention (National Institutes of Health, 2001).

A bone mineral density (BMD) test is used to measure the amount of calcium and other mineral types in the bone. BMD testing can help identify and diagnose osteoporosis, measure the likelihood of fractures, monitor the effectiveness of treatment for osteoporosis, and more (National Institute of Arthritis and Musculoskeletal and Skin Diseases, 2023). Currently, the dual-energy X-ray absorptiometry (DEXA) test is the most common method to measure bone mineral density (National Institute of Arthritis and Musculoskeletal and Skin Diseases, 2023). Although it is the preferred method of measuring bone mineral density, results are analyzed manually, and it is time consuming.

Computed Tomography (CT) scans are also commonly used for studying bones but are not often used for functional imaging or precise measurements. Instead, clinicians rely on DEXA scans because they're non-invasive and cost-effective (Jeong-Woon et al., 2023). CT scans are better at assessing bone density because it is the only scan that allows for the measurement of the real volume of density of the middle vertebrate. In a study of 140 participants, researchers found that only 17.1% of participants were diagnosed with osteoporosis using the DEXA test compared to 46.4% using quantitative computed tomography (QCT) scans (Li et al., 2013).). QCT scans refer to the analyses of CT images beyond a visual radiological evaluation in general radiology (Engelke et al., 2008). This shows there are effective methods to measure bone mineral density using CT scans rather than the DEXA scan.

The advancement in diagnostic tools has been imperative in managing osteoporosis and other bone-related conditions. The traditional method, dual-energy X-ray absorptiometry (DEXA), remains the gold standard for bone density assessments. However, its two-dimensional imaging capacity may not accurately depict the three-dimensional architecture of bones, leading to potential inaccuracies in fracture risk assessment and bone strength (Beckmann, 2023; Berger-Groch et al., 2020).

DEXA's limitations arise from its 2D nature, potentially misrepresenting bone integrity, especially in complex anatomical areas like the spine. This method also heavily relies on manual interpretation, which can be time-consuming and susceptible to human error, affecting consistency in patient diagnosis and management (Beckmann, 2023; Berger-Groch et al., 2020). Furthermore, it often requires a calibration phantom for every examination, and the radiation exposure is relatively high (van Hamersvelt et al., 2017; Mussmann et al., 2018).

To overcome these limitations, there is a shift towards using computed tomography (CT) scans, which offer three-dimensional imaging, providing a more comprehensive view of bone architecture. Recent studies have suggested the potential of using Hounsfield Unit (HU) values from CT scans as reliable indicators of bone density, capable of detecting subtle changes in bone quality that DEXA scans might miss (Lee et al., 2013). Several research teams have devised approaches to apply this principle. Numerous studies indicate that osteoporosis diagnosis is feasible using CT attenuation measured in Hounsfield units (HU), even with a single energy band (Pickhardt et al., 2011; Schreiber et al., 2011; Hendrickson et al., 2018). Typically, they designate a bone region of interest, calculate average HU values within that region for each subject, assess the correlation between HU values and reference DEXA T-scores, and identify HU thresholds corresponding to the T-score boundary between normal and abnormal values. While their methods are user-friendly, they are constrained to binary classification and do not furnish numerical BMD estimates to effectively aid clinicians (Jeong-Woon et al., 2023). In one study, it has been found that HU density in CT scans can be used in cases where DEXA could not be diagnosed, or DEXA measurements would not be possible in the detection of osteoporosis, but it would not be sufficient in detecting osteopenia (Ozturk et al., 2024).

Machine learning has found extensive application in various pharmaceutical domains, including diagnosis, imaging analysis, and treatment optimization. Notable examples include IBM Watson, which stands out as one of the most widely recognized platforms leveraging machine learning in healthcare (Erickson et al, 2017). The integration of machine learning algorithms with CT imaging represents a transformative advancement in medical imaging. Machine learning models can automate the segmentation and analysis of CT images, thereby reducing human error and increasing diagnostic speed. These models are trained to identify and analyze patterns in the imaging data that might indicate changes in bone density or structural weaknesses leading to potential fractures (Yousefi et al., 2019). Moreover, machine learning can seamlessly integrate diverse datasets as features simultaneously, regardless of their direct relevance to the disease, thereby enhancing the diagnostic reliability (Miura et al., 2024).

This novel approach could significantly change the landscape of bone health assessment by providing a more accurate, efficient, and non-invasive method to assess bone density and fracture risk. This would not only enhance patient outcomes through earlier and more precise detection but could also lead to the personalization of treatment plans, ultimately contributing to better management of conditions like osteoporosis (Berger-Groch et al., 2020). In recent times, machine learning models have played a significant role in speeding up CT scan image analysis. Moreover, it has been found that, AI inclusive methods such as deep learning (DL) is well suited as an auxiliary tool in clinical practice and as an automatic screener for identifying latent patients in computational tomography databases (Jeong-Woon et al., 2023).

Furthermore, machine learning can seamlessly integrate diverse data as features simultaneously, regardless of their direct relevance to the disease. This integration is anticipated to enhance the reliability of diagnosis (Miura et al., 2024). For instance, Monte-Moreno employed machine learning in the noninvasive estimation of blood glucose and pressure using photoplethysmography data. Their study demonstrated that variables like age and weight, which may not contribute significantly to prediction individually, can be combined with photoplethysmography data to enhance prediction accuracy (Monte-moreno et al., 2011). We have used a pixel density-based machine learning algorithm that may determine the bone density in a few steps. We want to establish this algorithm as an alternative to HU units in the future. Also, this model works on each segment separately so accuracy should be better than many other ML models in this field.

3. Connection to Explainability and HCI

For the ML agent to be successful, it must be able to team well with humans. In a high-risk industry like medicine, it is important for medical experts to feel comfortable using their domain knowledge to ensure the machine's predictions are accurate. The tool will use human-machine teaming methodologies to ensure humans are kept in the loop and are providing feedback to the tool to help improve its trustworthiness and explainability. By including medical experts in the process, it will help improve human trust with the ML agent. Medical professionals will be able to give feedback to the ML agent throughout its learning process, to allow for continuous learning and improved accuracy.

Medical experts can review labeled and segmented data using their expertise to evaluate the model's performance. They can provide feedback to the agent's predictions to make the algorithm's labeling and segmentation processes more efficient and improve its accuracy. Feedback can also be used to adjust the model's parameters to increase efficiency and accuracy of labeling, segmentation, and calculations.

In addition to labeling and segmentation, medical experts will be able to review the bone density and T-score outputs to determine if processes need to be updated to provide a more accurate reading. For transparency and to build human trust, the team has calculated confidence intervals of segments L1-L4. Medical professionals will be able to review these figures from the segments where the confidence intervals may be determined to be high based on their expertise within the field. The team has focused on increasing explainability and human trust of the algorithm by including comments in the code to explain how the algorithm works, using simple and well-known calculations found in the field of osteoporosis management, and giving medical personnel the ability to make updates to the algorithm as needed.

4. Methodology

Identifying spine and vertebral segments is critical in calculating the bone mineral density (BMD) of each segment of the spine. Bone mineral densities were calculated using segmented and labeled data from the VerSe '20 Large Scale Vertebrae Segmentation Challenge, example shown in Figure 1, consisting of 374 CT spine scans from 355 patients (Sekuboyina et al., 2021). However, demographic information and other information associated with patients were unknown. For this assignment, 30 CT scan images from 30 different patients were chosen, and it was assumed that the segmented and labeled information on the CT spine scans from the dataset were correct. Data from the training set was used to build on the existing algorithm. The algorithm captured segmented data at levels C1-C7, T1-T13, L1-L6, Sacrum, and Coccyx. One of the areas the traditional DEXA test focuses on to obtain bone mineral density is the lower lumbar spine (Marissa et al., 2018), L1-L4. To allow results from our algorithm and a DEXA test to be comparable in the future, we specifically focused on segments L1-L4.

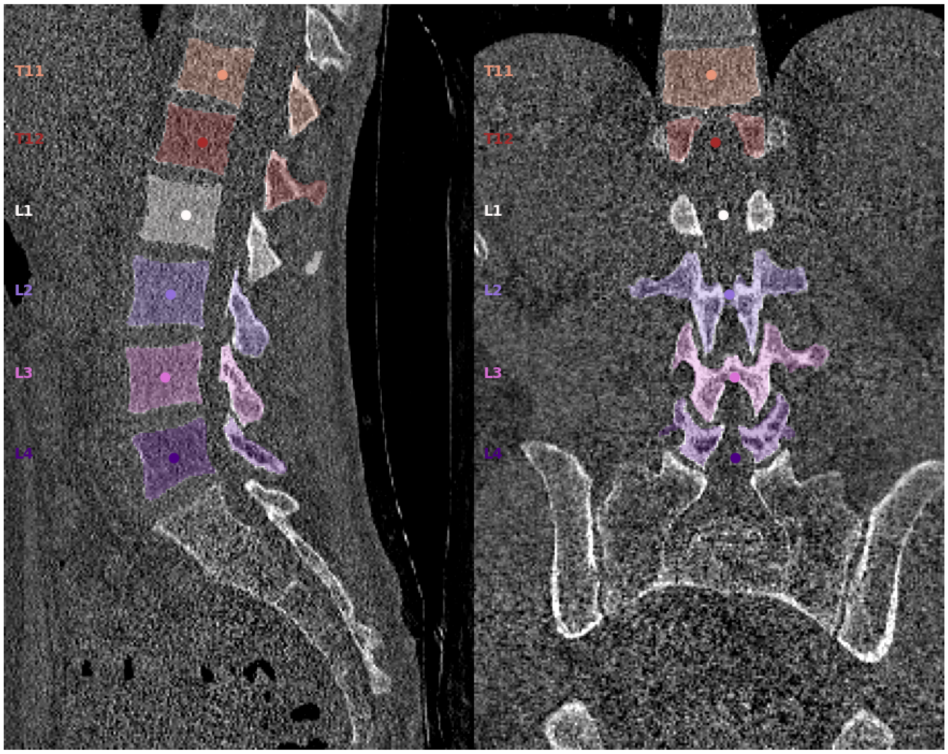

Figure 1. A CT scan which includes a segmented and label vertebral image from the data

The output of our machine learning algorithm was a Matplotlib plot, which was converted into an image array. The image array was then used as the basis for pixel density analysis. The bone pixel count of each segment was then divided by the bone area to determine its bone density. We built upon the provided VerSe Segmentation algorithm to not only include the calculation of BMD, but also, the calculated T-scores of the chosen patients Since CT scans are three-dimensional, we had the chance to express the average BMD in both gm/cm^3 and gm/cm^2. We used the bone area to figure out the average bone mineral density of that segment. Patient information was unknown in this dataset so we could not make a precise assumption based on the listed factors.

For the calculation of bone density, we created a for loop based on segments and counts in bone pixel count. The image array helped us count the pixels of a specific segment. We closed the for loop by defining the density as count/bone area and then printed out the results for each segment. We stored the results in Excel and analyzed that data set for further calculations and results. All the bone density values, in g/cm^2, were converted to T-scores for ease of understanding and comparison. A T-score was calculated for each patient to examine their bone health: T-score = (BMD − μ_{ref})/σ_{ref} (Frost et al., 2000), where μ_{ref} was the mean and σ_{ref} was the standard deviation of the population from the 30 patient CT scans (Kang et al., 2023). The bone mineral densities of L1-L4 of each patient were averaged and used as “BMD” in the T-score formula to assess the lower lumbar spine of the patient.

The T-score was calculated to determine and classify if a patient was considered to have "normal" or healthy bones, osteopenia (a less severe form of low bone mineral density than osteoporosis), or osteoporosis. A T-score represents how close a patient is to average peak bone density. If a patient's T-score was greater than or equal to –1, the algorithm identified their bones as healthy, and the patient was classified as "Normal" condition. If the patient's T-score was between –1 and –2.5, their bone density was low, and the patient's condition was classified as "Osteopenia". T-scores less than or equal to –2.5 were classified as "Osteoporosis" due to weak bone health (Kanis, 1994).

Confidence intervals were calculated to determine the stability of the predicted bone mineral densities. A 95% confidence interval was calculated using the average bone densities across patients of each segment. The calculation for confidence interval: $CI = \bar{x} \pm z\frac{s}{n}$, where $\bar{x}$ is the sample mean, z is the confidence level value, s is the sample standard deviation, and n is the sample size.

5. Results & Discussion

5. 1 Bone Health Categorization

Analysis from the diagnostics results of patient T-scores were synthesized to compare the patient diagnosis. Of the 30 patients, 2 patients (0.067%) were found to have osteopenia. It was concluded that most patients had normal/healthy bones shown in Figure 2.

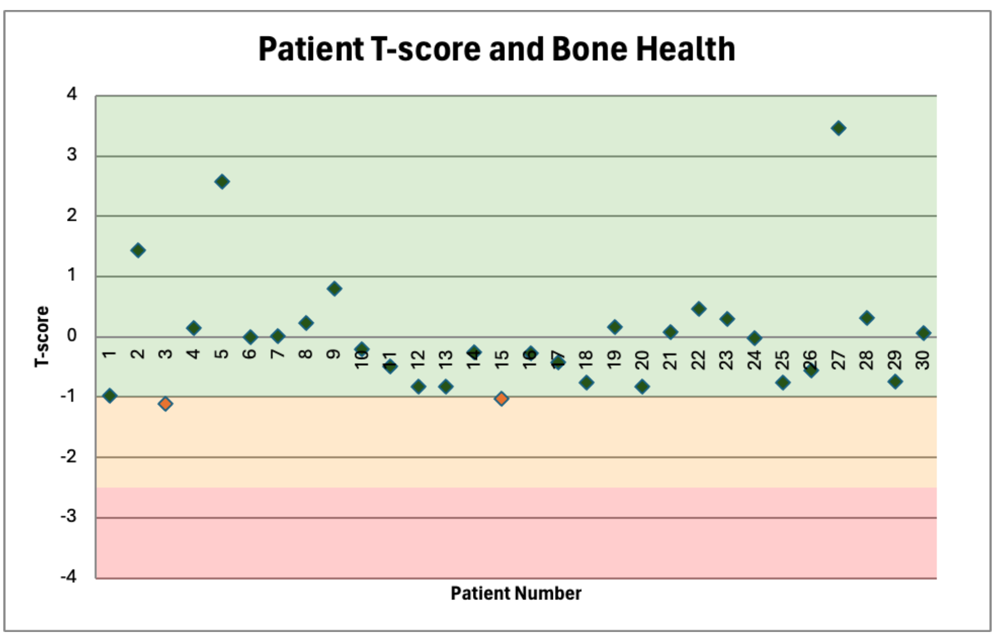

Figure 2. Graphic comparison of patient T-scores and their associated condition: Normal (green), Osteopenia (orange), or Osteoporosis (red).

5.2 Comparisons of L1-L4 Segments

Segment	n	Mean	Std. Dev.	95% CI	CI (-)	CI (+)
L1	30	1.8397	1.0327	0.3695	1.4702	2.2093
L2	30	1.9017	1.0629	0.3804	1.5213	2.2820
L3	30	1.8671	1.1396	0.4078	1.4593	2.2749
L4	30	1.8773	1.1766	0.4210	1.4563	2.2984
L1 to L4Avg	30	1.8715	1.1047	0.3953	1.4762	2.2668

Table 1. Comparison between the different segments of the lower lumbar spine. Metrics include population mean and standard deviation, and confidence intervals. Units in (g/ cm^2).

The table provides a statistical comparison of the L1-L4 lumbar spine segments using a sample size of 30 for each segment. Mean values ranged from 1.8397 g/ cm^2 for L1 to 1.8773 g/ cm^2 for L4, indicating little variability in measurements within these segments. The standard deviations are constant and range from 1.03 to 1.17, indicating relatively similar measurement variance around the mean for each segment. The 95% confidence intervals represent the region within which the true mean can fall, suggesting higher accuracy in segments L3 and L4, marked by narrower confidence intervals (CI(-) and CI(+)) compared to L1 and L2. These data suggest that there is a pattern or trend in bone density or structural measurements from the upper to lower lumbar segments, with the lower segments (L3 and L4) showing slightly higher mean values and more accurate estimates. Such findings may lead to further investigation of the physiological reasons for this distribution and assessment of whether these differences are clinically significant or the result of natural variations in human anatomy.

5.3 Bone Densities within Segments

We also compared the bone density data among various segments of the same CT scans to see the similarities or differences in bone segments or not. The maximum and minimum bone densities of segments L1-L4 were evaluated to get more information on the patient as shown in Table 2. Although the focus of this research is to predict and diagnose osteoporosis, which is developed when there is a decrease in bone mineral density, very high bone mineral densities can be problematic and a result of degenerative diseases or underlying disorders affecting the skeleton (Gregson et al., 2013).

In Figure 3, we can see most of the patients have similar patterns in bone density from L1-L4. In most cases, L2 and L4 seem to have comparatively higher BMD values than L3 and L1. We have also converted all the bone density numbers to respective T-scores from the equation T-score = (BMD − μref)/σref (Frost et al., 2000), where μref was the mean and σref was the standard deviation of the population from the 30 patient CT scans (Kang et al., 2023). The results are shown in the graphic comparison below.

Segment	n	Maximum Bone Density	Minimum Bone Density
L1	30	5.5230	0.6100
L2	30	5.6580	0.6830
L3	30	5.7450	0.6300

L4	30	5.8100	0.6420

Table 2. Comparison of the maximum and minimum population bone densities in and between segments L1-L4. Units in (g/ cm^2).

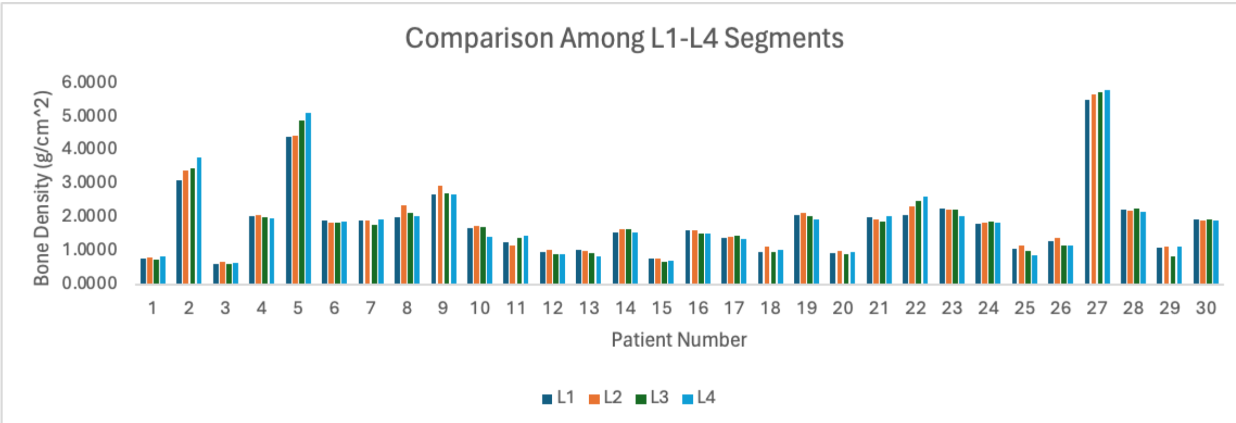

Figure 3. Comparison of L1-L4 segments for each patient

5.4 Discussion

VerSe '20 Large Scale Vertebrae Segmentation Challenge dataset had several limitations. Firstly, the data contains older data and there is not enough information about the patients. This barrier hindered feature selection when building out and updating the algorithm. Secondly, our model is currently a simple tool for pixel density measurements. We did not train, test or validate our model due to time constraints. Although there were 374 CT spine scans from 355 patients, we only used 30 CT scans from 30 patients in the training set. If all the data given was incorporated in our algorithm, the results would potentially change and look different. For future research, we would want to compare the algorithm's results with actual T-scores from the DEXA tests, using the same data, as well as with manual observations from expert radiologists. Moreover, we can compare the results with results collected via other machine learning models in this field such as Artificial Neural Networks (ANN) and Convoluted Neural Networks (CNN).

6. Conclusion

This research focuses on applying a straightforward machine learning model to the field of spine research. The model predicts bone mineral density in g/cm^2 and converts that to the T-score and identifies osteoporotic vertebrae through the analysis of the pixel density from conventional segmented CT scans. This aids spinal surgeons in avoiding underestimation of osteoporosis in the spine before surgery. With a larger dataset, we anticipate that the predictive accuracy of our model will improve even further. By increasing the size of this dataset, we can enhance our algorithms and possibly uncover subtle patterns that would not be visible with smaller data. For instance, other factors including age, gender, and medical history may impact the T-score. Potential future research is to create a more powerful model by merging these variables, which will enable us to enhance prediction accuracy and tailor treatment regimens. Our model's clinical utility will be greatly enhanced by this development, making it a vital tool for patient care plans that are tailored to the individual and diagnostically active practice. We would also like to validate our dataset and compare that with established bone density measurement techniques such as DEXA scans and some other machine learning models that use Artificial Neural Networks and Deep learning.

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Author contributions have been recorded below. Each author attended working meetings and contributed to formalizing thoughts, insights, and the writing of this report. In these meetings, the team worked together to discuss and determine the problem to be addressed, discuss the direction and problems with of the algorithm, gather information, outline the document, and develop the structure of each section.

Each author contributed equally. The authors confirm contribution to the paper as follows: study conception and design: Deborah Asabere (DA), Mashnur Rashid (MR), Yerdigul Uatkhan (YU); data collection: DA, MR, YU; analysis and interpretation of results: DA, MR, YU; draft manuscript preparation: DA, MR, YU; team presentation: DA, MR, YU. All authors reviewed the results and approved the final version of the manuscript.

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